

Отчёт о лабораторной работе

Лабораторная работа 3

Мантуров Татархан Бесланович

Содержание

Цель работы

Изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.

Выполнение

Пропишем команду `ipconfig`. В качестве вывода мы получим различные соединения, в том числе наше активное (последнее). Из получаемой информации видим ipv6 адрес, ipv4 адрес, маску подсети и основной шлюз (рис. [-@fig:001]).

```
C:\Users\taotel>ipconfig

Настройка протокола IP для Windows

Адаптер Ethernet Ethernet:

    Состояние среды. . . . . : Среда передачи недоступна.
    DNS-суффикс подключения . . . . : Dlink

Адаптер Ethernet Ethernet 3:

    DNS-суффикс подключения . . . . :
    Локальный IPv6-адрес канала . . : fe80::6144:d07b:6a92:c140%8
    IPv4-адрес. . . . . : 192.168.56.1
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз. . . . . :

Адаптер Ethernet Ethernet 5:

    DNS-суффикс подключения . . . . :
    Локальный IPv6-адрес канала . . : fe80::e1d9:3db8:b158:6615%6
    IPv4-адрес. . . . . : 192.168.1.1
    Маска подсети . . . . . : 255.255.255.0
    Основной шлюз. . . . . :
```

`ipconfig`

Указав к `ipconfig` ключ `/all`, мы получим более подробную информацию, в том числе физический адрес (mac). В нашем случае mac адрес имеет вид 74-04-f1-d0-d0-45. Первые 3 байта - это идентификатор производителя, а остальные 3 - интерфейса. В первом байте (74) зашита информация о том, что адрес индивидуальный, и что является глобально администрируемым, поскольку последние 2 бита 74 в двоичной системе, если переводить из hex, равны нулям (рис. [-@fig:002]).

```
C:\Users\taotel>ipconfig /all

Настройка протокола IP для Windows

Имя компьютера . . . . . : DESKTOP-R72KLSN
Основной DNS-суффикс . . . . . :
Тип узла . . . . . : Гибридный
IP-маршрутизация включена . . . . . : Нет
WINS-прокси включен . . . . . : Нет

Адаптер Ethernet Ethernet:

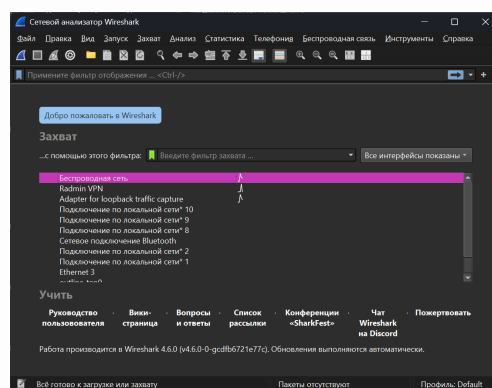
Состояние среды . . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . : Dlink
Описание . . . . . : Realtek PCIe GbE Family Controller
Физический адрес . . . . . : 04-7C-16-35-7A-DE
DHCP включен . . . . . : Да
Автонастройка включена . . . . . : Да

Адаптер Ethernet Ethernet 3:

DNS-суффикс подключения . . . . . :
Описание . . . . . : VirtualBox Host-Only Ethernet Adapter
Физический адрес . . . . . : 0A-00-27-00-00-08
DHCP включен . . . . . : Нет
Автонастройка включена . . . . . : Да
```

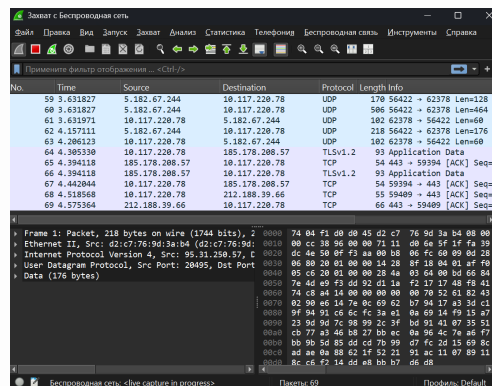
ipconfig /all

Откроем Wireshark и выберем сетевое подключение, которое активно. В нашем случае, это беспроводная сеть (рис. [-@fig:003]).



Выбор соединения

Так выглядит сам интерфейс (рис. [-@fig:004]).



Wireshark

Попробуем пингануться к шлюзу, адрес которого мы получили в самом начале (рис. [-@fig:005]).

```
C:\Users\taotel>ping 8.8.8.8

Обмен пакетами с 8.8.8.8 по 32 байтами данных:
Ответ от 8.8.8.8: число байт=32 время=22мс TTL=108
Ответ от 8.8.8.8: число байт=32 время=22мс TTL=108
Ответ от 8.8.8.8: число байт=32 время=22мс TTL=108
Ответ от 8.8.8.8: число байт=32 время=22мс TTL=108
```

ping

Переходим обратно в Wireshark и посмотрим на ICMP пакеты. Поскольку было передано 4 пакета, то мы видим 8 записей (на каждый пакет получение и отправка) (рис. [-@fig:006]).

No.	Time	Source	Destination	Protocol	Length	Info
472	68.836280	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=86
473	68.837234	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=86
475	61.840797	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=87
476	61.864719	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=87
494	62.862111	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=88
495	62.868768	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=88
580	63.876617	10.117.220.78	10.117.220.134	ICMP	74	Echo (ping) request id=0x0001, seq=89
581	63.890386	10.117.220.134	10.117.220.78	ICMP	74	Echo (ping) reply id=0x0001, seq=89

icmp

Изучим самый первый, исходящий пакет. Как видим, длина фрейма составляет 74 байта, используется тип ethernet 2 и мы так же можем видеть физические адреса отправителя (то есть наш) и получателя (то есть шлюза). Оба адреса являются индивидуальными и глобально администрируемыми (рис. [-@fig:007]).

```

Epoch Arrival Time: 1768187218.767445000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 866.353000 milliseconds]
[Time since reference or first frame: 1 minute, 856.283000 milliseconds]
Frame Number: 472
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
  = Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
    .....0 ..... = 10 bit: Locally administered address (this is NOT the factory default)
    .....0 ..... = 10 bit: Individual address (unicast)
  = Source: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1)
    .....0 ..... = 10 bit: Individual address (unicast)
    .....0 ..... = 10 bit: Globally unique address (factory default)
    .....1 ..... = 10 bit: Locally administered address (this is NOT the factory default)
Type: IPv4 (0x0000)
(Stream index: 0)
  
```

Анализ исходящего пакета

Теперь посмотрим на ответный фрейм. Он, по сути, практически не отличается от исходящего, только адреса отправителя и получателя поменяны местами (рис. [-@fig:008]).

```

Frame 473: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{87D3937E-0B51-498A-B35E-C81731C8CFA}
Section number 1
Interface 0: \Device\NPF_{87D3937E-0B51-498A-B35E-C81731C8CFA}
Encapsulation type: Ethernet (1)
Arrival Time: Oct 11, 2025 12:53:38.788477000 RTT: 2 (same)
UTC Arrival Time: Oct 11, 2025 12:53:38.788477000 UTC
Epoch Arrival Time: 1768187218.788477000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 21.032000 milliseconds]
[Time delta from previous displayed frame: 21.032000 milliseconds]
[Time since reference or first frame: 1 minute, 857.234000 milliseconds]
Frame Number: 473
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:icmp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: ICMP]
[Coloring Rule String: icmp || icmpv6]
Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1)
  = Destination: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1)
    .....0 ..... = 10 bit: Globally unique address (factory default)
    .....0 ..... = 10 bit: Individual address (unicast)
  = Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
    .....1 ..... = 10 bit: Locally administered address (this is NOT the factory default)
    .....0 ..... = 10 bit: Individual address (unicast)
Type: IPv4 (0x0000)
(Stream index: 0)
  
```

Анализ входящего пакета

Посмотрим теперь на ARP пакеты. Как видим, в них не так много информации. Они отвечают за получение физического адреса. Так, мы видим, что первый ARP пакет пытается получить адрес нашего устройства (рис. [-@fig:009]).

```

Frame 165: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{87D3937E-0B51-498A-B35E-C81731C8CFA}
Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dst: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1)
  = Destination: Intel_08:00:45:74:04:F1 (08:00:45:74:04:F1)
    .....0 ..... = 10 bit: Globally unique address (factory default)
    .....0 ..... = 10 bit: Individual address (unicast)
  = Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)
    .....0 ..... = 10 bit: Locally administered address (this is NOT the factory default)
    .....0 ..... = 10 bit: Individual address (unicast)
Type: ARP (0x0008)
(Stream index: 0)
  00:10:00:00:00:00, 00:10:00:00:00:00, 00:10:00:00:00:00
  Address Resolution Protocol (request)
  
```

arp

Второй арп пакет является обратным, то есть мы получаем адрес шлюза. В обоих случаях в заголовке ethernet 2 хранится физический адрес обоих устройств, но в первом случае тип является vlan, а во втором - арп (рис. [-@fig:010]).

No.	Time	Source	Destination	Protocol	Length	Info
165	25.988466	d2:c7:76:9d:3a:b4	Intel_08:00:45	ARP	42	Who has 10.117.220.78? Tell 10.117.2
166	25.988503	Intel_08:00:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:00:00:4
167	25.988540	d2:c7:76:9d:3a:b4	Intel_08:00:45	ARP	42	Who has 10.117.220.78? Tell 10.117.2
168	25.988577	Intel_08:00:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:00:00:4
169	25.988614	d2:c7:76:9d:3a:b4	Intel_08:00:45	ARP	42	Who has 10.117.220.78? Tell 10.117.2
170	25.988651	Intel_08:00:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:00:00:4

Frame 166: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{B7D3937E-83} Ethernet II, Src: Intel_08:00:45 (74:04:f1:00:00:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

.....0..... = 48 bit: Locally administered address (this is NOT the factory default)

.....0..... = 30 bit: Individual address (unicast)

.....0..... = 48 bit: Locally administered address (factory default)

.....0..... = 30 bit: Individual address (unicast)

Type: ARP (8000)

[Stream index: 0]

Address Resolution Protocol (reply)

Второй арп пакет

Теперь пинганём яндекс (рис. [-@fig:011]).

```
C:\Users\taote1>ping rudn.ru
ping ya.ru
```

Посмотрим на арп пакеты. Как видим, их 2. Как и в прошлый раз, клиент и шлюз пытаются обменяться физическими адресами (рис. [-@fig:012]).

No.	Time	Source	Destination	Protocol	Length	Info
50	8.723442	d2:c7:76:9d:3a:b4	Intel_08:00:45	ARP	42	Who has 10.117.220.78? Tell 10.117.2
51	8.723487	Intel_08:00:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:00:00:4

Frame 50: Packet, 46 bytes on wire (368 bits), 46 bytes captured (368 bits) on interface \Device\NPF_{B7D3937E-83} Ethernet II, Src: Intel_08:00:45 (74:04:f1:00:00:45), Dst: Intel_08:00:45 (74:04:f1:00:00:45)

Destination: Intel_08:00:45 (74:04:f1:00:00:45)

.....0..... = 48 bit: Locally administered address (factory default)

.....0..... = 30 bit: Individual address (unicast)

Source: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

.....0..... = 48 bit: Locally administered address (this is NOT the factory default)

.....0..... = 30 bit: Individual address (unicast)

Type: ARP (8000)

[Stream index: 0]

802.1Q Virtual LAN (Vlan200)

Address Resolution Protocol (request)

арп пакеты

Второй арп пакет отличается, как обычно, только направлением (рис. [-@fig:013]).

No.	Time	Source	Destination	Protocol	Length	Info
50	8.723442	d2:c7:76:9d:3a:b4	Intel_08:00:45	ARP	42	Who has 10.117.220.78? Tell 10.117.2
51	8.723487	Intel_08:00:45	d2:c7:76:9d:3a:b4	ARP	42	10.117.220.78 is at 74:04:f1:00:00:4

Frame 51: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{B7D3937E-83} Ethernet II, Src: Intel_08:00:45 (74:04:f1:00:00:45), Dst: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

Destination: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4)

.....0..... = 48 bit: Locally administered address (this is NOT the factory default)

.....0..... = 30 bit: Individual address (unicast)

Source: Intel_08:00:45 (74:04:f1:00:00:45)

.....0..... = 48 bit: Locally administered address (factory default)

.....0..... = 30 bit: Individual address (unicast)

Type: ARP (8000)

[Stream index: 0]

Address Resolution Protocol (reply)

Второй arg пакет

Теперь посмотрим на істр пакеты. их как обычно 8. Как видим, тут уже есть небольшая разница. Теперь мы обращаемся не к шлюзу, а прямо к серверу яндекса. Мы можем видеть, что ір другой. Однако, в рамках ethernet 2 мы всё ещё общаемся со шлюзом (рис. [-@fig:014]).

```

No.    Time           Source                Destination            Protocol Length Info
-----
24 3.422100      10.117.220.78        77.88.44.242          ICMP 74 Echo (ping) request IIdwv0001, seq=...
25 3.468000      77.88.44.242         10.117.220.78         ICMP 74 Echo (ping) reply IIdwv0001, seq=...
26 4.438600      10.117.220.78        77.88.44.242          ICMP 74 Echo (ping) request IIdwv0001, seq=...
29 4.477100      77.88.44.242         10.117.220.78         ICMP 74 Echo (ping) reply IIdwv0001, seq=...
31 5.555200      10.117.220.78        77.88.44.242          ICMP 74 Echo (ping) request IIdwv0001, seq=...
43 5.562100      77.88.44.242         10.117.220.78         ICMP 74 Echo (ping) reply IIdwv0001, seq=...
45 5.562100      10.117.220.78        77.88.44.242          ICMP 74 Echo (ping) request IIdwv0001, seq=...
46 5.529173      77.88.44.242         10.117.220.78         ICMP 74 Echo (ping) reply IIdwv0001, seq=...

Frame 24: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface DeviceNPF_{B7D31937E-...}
Sequence number: 1
+ Interface ID: 0 (DeviceNPF_{B7D31937E-8951-458A-B3C8-CE1731FC6FAF})
+ Encapsulation type: Ethernet [15]
Arrival time: Oct 11, 2025 16:02:45.176400000 RTT: 2 (same)
UTC Arrival Time: Oct 11, 2025 15:02:45.176400000 UTC
Epoch Arrival Time: 1768188778.176400000
[Time shift for this packet: 0.000000000 seconds]
Time delta from previous captured frame: 31.5550000 milliseconds
[Time since reference or first frame: 3.422000000 seconds]
Frame Number: 24
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked:] False
[Frame is ignored:] False
[Protocols in frame show ethertype: ip:icmp:data]
Character encoding: ASCII (0)
Coloring Rule Name: None
[Coloring Rule String: icmp || icmpseq]
+ ethernet II, Src: Intel_08:00:45:76:8d:43, Dst: d2:c7:76:9d:b4:d2 (c7:76:9d:b4:d2)
+ Destination: d2:c7:76:9d:b4:d2 (c7:76:9d:b4:d2)
+ ... .. . . . . . . . . . . L0 Bitt (locally administered address (this is NOT the factory default))
+ Source: 10.117.220.78 (10.117.220.78)
+ ... .. . . . . . . . . . . H0 Bitt (Internally assigned (uncast))

```

Исходящий пакет istr

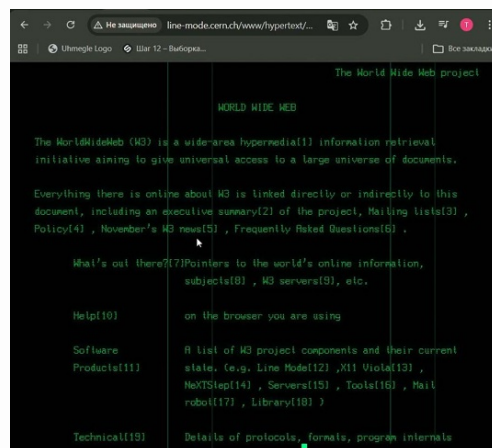
Входящий пакет *istr*, то есть, ответный, не будет сильно отличаться от входящего, только направлением (рис. [-@fig:015]).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 0)
2	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 0)
3	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 1)
4	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 1)
5	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 2)
6	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 2)
7	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 3)
8	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 3)
9	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 4)
10	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 4)
11	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 5)
12	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 5)
13	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 6)
14	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 6)
15	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 7)
16	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 7)
17	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 8)
18	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 8)
19	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 9)
20	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 9)
21	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 10)
22	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 10)
23	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 11)
24	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 11)
25	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 12)
26	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 12)
27	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 13)
28	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 13)
29	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 14)
30	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 14)
31	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 15)
32	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 15)
33	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 16)
34	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 16)
35	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 17)
36	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 17)
37	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 18)
38	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 18)
39	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 19)
40	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 19)
41	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 20)
42	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 20)
43	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 21)
44	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 21)
45	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 22)
46	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 22)
47	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 23)
48	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 23)
49	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 24)
50	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 24)
51	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 25)
52	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 25)
53	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 26)
54	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 26)
55	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 27)
56	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 27)
57	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 28)
58	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 28)
59	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 29)
60	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 29)
61	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 30)
62	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 30)
63	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 31)
64	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 31)
65	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 32)
66	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 32)
67	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 33)
68	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 33)
69	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 34)
70	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 34)
71	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 35)
72	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 35)
73	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 36)
74	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 36)
75	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 37)
76	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 37)
77	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 38)
78	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 38)
79	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 39)
80	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 39)
81	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 40)
82	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 40)
83	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 41)
84	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 41)
85	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 42)
86	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 42)
87	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 43)
88	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 43)
89	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 44)
90	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 44)
91	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 45)
92	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 45)
93	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 46)
94	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 46)
95	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 47)
96	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 47)
97	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 48)
98	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 48)
99	0.000000	192.168.1.100	192.168.1.1	ICMP	60	echo (request id 49)
100	0.000000	192.168.1.1	192.168.1.100	ICMP	60	echo (reply id 49)

```
Frame 5: Packet 14, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface Dvnic0\NPF{803B2070-0015-408A-8035-CB7353C0FA}, id 0
    Ethernet II, Src: Dvnic0\NPF{803B2070-0015-408A-8035-CB7353C0FA}, Dst: Dvnic0\NPF{803B2070-0015-408A-8035-CB7353C0FA}, Protocol: ICMP (1)
    Internet Control Message Protocol (ICMP)
        Echo (request id 14)
            0x0000: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0010: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0050: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0060: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0080: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0090: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x00F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0100: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0110: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0120: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0130: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0140: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0150: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0160: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0170: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0180: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0190: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x01F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0200: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0210: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0220: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0230: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0240: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0250: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0260: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0270: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0280: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0290: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x02F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0300: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0310: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0320: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0330: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0340: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0350: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0360: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0370: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0380: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0390: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03A0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03B0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03C0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03D0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03E0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x03F0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
            0x0400: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 
```

Входящий пакет істр

Посетим сайт по протоколу http (рис. [-@fig:016]).



Подключение по http

Посмотрим на http пакеты. Как видим, они есть и входящие, и исходящие. Работает http поверх tcp. Мы видим, что информация в разделе о tcp говорит о входящем и исходящем портах, о длине сегмента, флагах и чексумме (рис. [-@fig:017]).

The image shows a Wireshark packet capture of an outgoing HTTP packet. The packet list at the top shows packet 88, which is an HTTP GET request to 188.184.67.127. The packet details pane shows the following information:

- Source Port: 56283
- Destination Port: 80
- Stream Index: 42
- Stream Packet Number: 4
- Conversation completeness: Complete, WITH_DATA (31)
- TCP Segment Len: 361
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 235742865
- Next Sequence Number: 395 (relative sequence number)
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment Number (raw): 235944299
- 0001> Header Length: 20 bytes (5)
- Flags: 0001 (PSH, ACK)
- Window: 355
- [Calculated window size: 65280]
- Window size scaling factor: 255
- Checksum: 00e82 [unverified]
- Checksum Status: Unverified
- Urgent Pointer: 0
- [Timestamps]
- Time since first frame in this TCP stream: 158.809000 milliseconds
- Time since previous frame in this TCP stream: 355.000 microseconds
- [SQ/ACK analysis]

Исходящий http пакет

Входящий пакет же будет отличаться тем, что длина сегмента и чексумма будут другие, а порты поменяются местами (рис. [-@fig:018]).

The image shows a Wireshark packet capture of an incoming HTTP packet. The packet list at the top shows packet 94, which is an HTTP 200 OK response from 188.184.67.127. The packet details pane shows the following information:

- Source Port: 80
- Destination Port: 56283
- Stream Index: 22
- Stream Packet Number: 6
- Conversation completeness: Complete, WITH_DATA (31)
- TCP Segment Len: 878
- Sequence Number: 1 (relative sequence number)
- Sequence Number (raw): 315864299
- Next Sequence Number: 879 (relative sequence number)
- Acknowledgment Number: 382 (relative ack number)
- Acknowledgment Number (raw): 235743346
- 0001 ...> Header Length: 20 bytes (5)
- Flags: 0000 (PSH, ACK)
- Window: 240
- [Calculated window size: 31872]
- Window size scaling factor: 120
- Checksum: 00c1b [unverified]
- Checksum Status: Unverified
- Urgent Pointer: 0
- [Timestamps]
- [SQ/ACK analysis]
- Client Contiguous Streams: 1
- Server Contiguous Streams: 1
- TCP payload (878 bytes)

Входящий http пакет

Теперь рассмотрим dns пакеты. Они уже работают по udp. Сам раздел по udp даёт не так много информации, как tcp, однако информация о портах, длине и чексумме присутствует (рис. [-@fig:019]).

No.	Time	Source	Destination	Protocol	Length	Info
3	0.808996	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x8466 A info.cern.ch
4	1.020694	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x8466 A info.cern.ch
5	1.010404	10.117.220.134	10.117.220.78	DNS	112	Standard query response 0x8466 A info.cern.ch
6	1.115179	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x8704 AAAA weba.f902.c
7	1.119248	10.117.220.134	10.117.220.78	DNS	93	Standard query response 0x8704 A weba.f902.c
8	1.119567	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA weba.f902.c
9	1.362600	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA weba.f902.c
10	1.483479	10.117.220.134	10.117.220.78	DNS	185	Standard query response 0x8704 AAAA weba.f902.c

Frame 7: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface Device\NPF_{87D3937E-035E-4B42-80D0-000000000000} (0.0.0.0) Ethernet II, Src: Intel_08:00:00:00:00:00, Dst: 02:c7:76:9d:3a:b4 (02:c7:76:9d:3a:b4) Internet Protocol Version 4, Src: 10.117.220.78, Dst: 10.117.220.134 User Datagram Protocol, Src Port: 61668, Dst Port: 53 Source Port: 61668 Destination Port: 53 Length: 49 Checksum: 0xcdfb [unverified] [Checksum Status: Unverified] [Stream index: 1] [Stream Packet Number: 1] [Timestamps] UDP payload (35 bytes) Domain Name System (query)

Исходящий dns пакет

Входящий пакет с точки зрения udr будет отличаться поменянными местами портами, чексуммой и длиной (рис. [-@fig:020]).

No.	Time	Source	Destination	Protocol	Length	Info
3	0.808996	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x8466 A info.cern.ch
4	1.020694	10.117.220.78	10.117.220.134	DNS	72	Standard query 0x8466 A info.cern.ch
5	1.010404	10.117.220.134	10.117.220.78	DNS	112	Standard query response 0x8466 A info.cern.ch
6	1.115179	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA weba.f902.c
7	1.119248	10.117.220.134	10.117.220.78	DNS	93	Standard query response 0x8704 A weba.f902.c
8	1.119567	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA weba.f902.c
9	1.362600	10.117.220.78	10.117.220.134	DNS	77	Standard query 0x8704 AAAA weba.f902.c
10	1.483479	10.117.220.134	10.117.220.78	DNS	185	Standard query response 0x8704 AAAA weba.f902.c

Frame 8: Packet, 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface Device\NPF_{87D3937E-035E-4B42-80D0-000000000000} (0.0.0.0) Ethernet II, Src: 02:c7:76:9d:3a:b4 (02:c7:76:9d:3a:b4), Dst: Intel_08:00:00:00:00:00 (08:00:00:00:00:00) Internet Protocol Version 4, Src: 10.117.220.134, Dst: 10.117.220.78 User Datagram Protocol, Src Port: 53, Dst Port: 61668 Source Port: 53 Destination Port: 61668 Length: 59 Checksum: 0x55f1 [unverified] [Checksum Status: Unverified] [Stream index: 1] [Stream Packet Number: 2] [Timestamps] UDP payload (35 bytes) Domain Name System (response)
--

Входящий dns пакет

Теперь рассмотрим quic. Он работает поверх udr, и в рамках уровня udr содержит примерно ту же информацию, что и dns (рис. [-@fig:021]).

No.	Time	Source	Destination	Protocol	Length	Info
6762	719.471867	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KPR), DCID=0x0
6763	719.495981	104.76.24.37	10.117.220.78	QUIC	66	Protected Payload (KPR)
6764	719.507133	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (KPR), DCID=0x0
6765	719.511381	104.76.24.37	10.117.220.78	QUIC	1208	Protected Payload (KPR)
6766	719.511381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KPR)
6767	719.511381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KPR)
6768	719.511381	104.76.24.37	10.117.220.78	QUIC	283	Protected Payload (KPR)
6769	719.511381	104.76.24.37	10.117.220.78	QUIC	285	Protected Payload (KPR)
6770	719.511381	104.76.24.37	10.117.220.78	QUIC	78	Protected Payload (KPR)
6771	719.511903	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KPR), DCID=0x0
6772	719.512140	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KPR), DCID=0x0
6773	719.512497	10.117.220.78	104.76.24.37	QUIC	77	Protected Payload (KPR), DCID=0x0
6774	719.513886	104.76.24.37	10.117.220.78	QUIC	1404	Protected Payload (KPR)
6775	719.515387	10.117.220.78	104.76.24.37	QUIC	75	Protected Payload (KPR), DCID=0x0
6776	719.517815	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (KPR)
6777	719.517815	104.76.24.37	10.117.220.78	QUIC	1072	Protected Payload (KPR)
6778	719.517815	104.76.24.37	10.117.220.78	QUIC	77	Protected Payload (KPR)
6779	719.517815	104.76.24.37	10.117.220.78	QUIC	64	Protected Payload (KPR)
6780	719.519218	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KPR), DCID=0x0
6781	719.519383	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KPR), DCID=0x0
6782	719.519464	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (KPR), DCID=0x0
6783	719.519527	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (KPR), DCID=0x0
6784	719.519555	104.76.24.37	10.117.220.78	QUIC	1414	Protected Payload (KPR)
6785	719.519455	104.76.24.37	10.117.220.78	QUIC	875	Protected Payload (KPR)
6786	719.519520	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (KPR), DCID=0x0

Frame 6786: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface Device\NPF_{87D3937E-035E-4B42-80D0-000000000000} (0.0.0.0) Ethernet II, Src: Intel_08:00:00:00:00:00, Dst: 02:c7:76:9d:3a:b4 (02:c7:76:9d:3a:b4) Internet Protocol Version 4, Src: 10.117.220.78, Dst: 104.76.24.37 User Datagram Protocol, Src Port: 65113, Dst Port: 443 Source Port: 65113 Destination Port: 443 Length: 39 Checksum: 0x676d [unverified] [Checksum Status: Unverified] [Stream index: 102] [Stream Packet Number: 43] [Timestamps] UDP payload (31 bytes) QUIC (ICP)

Исходящий quic пакет

И различия между входящим и исходящим пакетом ровно такие же (рис. [-@fig:022]).

No.	Time	Source	Destination	Protocol	Length	Info
6762	719.471867	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (x90), DCID=0x0
6763	719.469581	104.76.24.37	10.117.220.78	QUIC	68	Protected Payload (x90)
6764	719.507133	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (x90), DCID=0x0
6765	719.531381	104.76.24.37	10.117.220.78	QUIC	1288	Protected Payload (x90)
6766	719.531381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (x90)
6767	719.531381	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (x90)
6768	719.531381	104.76.24.37	10.117.220.78	QUIC	383	Protected Payload (x90)
6769	719.531381	104.76.24.37	10.117.220.78	QUIC	185	Protected Payload (x90)
6770	719.531381	104.76.24.37	10.117.220.78	QUIC	78	Protected Payload (x90)
6771	719.531903	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (x90), DCID=0x0
6772	719.532140	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (x90), DCID=0x0
6773	719.532407	10.117.220.78	104.76.24.37	QUIC	77	Protected Payload (x90), DCID=0x0
6774	719.533806	104.76.24.37	10.117.220.78	QUIC	1494	Protected Payload (x90)
6775	719.535387	10.117.220.78	104.76.24.37	QUIC	75	Protected Payload (x90), DCID=0x0
6776	719.537815	104.76.24.37	10.117.220.78	QUIC	1292	Protected Payload (x90)
6777	719.537815	104.76.24.37	10.117.220.78	QUIC	1872	Protected Payload (x90)
6778	719.537815	104.76.24.37	10.117.220.78	QUIC	77	Protected Payload (x90)
6779	719.537815	104.76.24.37	10.117.220.78	QUIC	64	Protected Payload (x90)
6780	719.539210	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (x90), DCID=0x0
6781	719.539383	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (x90), DCID=0x0
6782	719.539464	10.117.220.78	104.76.24.37	QUIC	76	Protected Payload (x90), DCID=0x0
6783	719.539527	10.117.220.78	104.76.24.37	QUIC	74	Protected Payload (x90), DCID=0x0
6784	719.554855	104.76.24.37	10.117.220.78	QUIC	1494	Protected Payload (x90)
6785	719.554855	104.76.24.37	10.117.220.78	QUIC	73	Protected Payload (x90)
6786	719.559230	10.117.220.78	104.76.24.37	QUIC	73	Protected Payload (x90), DCID=0x0

Frame 6784: Packet, 1494 bytes on wire (11952 bits), 1494 bytes captured (11952 bits) on interface vDevice\NPF_{0...}

Ethernet II, Src: d2:c7:76:9d:3a:b4 (d2:c7:76:9d:3a:b4), Dat: Intel_00:00:45 (74:04:f1:00:00:45)

Internet Protocol Version 4, Src: 104.76.24.37, Dest: 10.117.220.78

User Datagram Protocol, Src Port: 443, Dest Port: 65115

Source Port: 443
Destination Port: 65115
Length: 1494
Checksum: 0x02a [unverified]
[Checksum Status: Unverified]
[Stream Index: 142]
[Stream Packet Number: 41]
[Timestamp]
UDP payload (1452 bytes)

QUIC RTT

Входящий quic пакет

Теперь рассмотрим процесс хендшейка. Для этого перезагрузим http страницу и посмотрим, что происходит перед тем, как происходит http запрос get. Для начала, клиент отправляет серверу syn запрос, что мы можем видеть во флагах tcp (рис. [-@fig:023]).

No.	Time	Source	Destination	Protocol	Length	Info
0	0.708788	10.117.220.78	188.184.67.127	TCP	60	56449 → 80 [SYN] Seq=0 Win=65535 Len=0
1	0.965410	10.117.220.78	188.184.67.127	TCP	60	56449 → 80 [SYN] Seq=0 Win=65535 Len=0
2	0.965918	188.184.67.127	10.117.220.78	TCP	60	80 → 56449 [SYN, ACK] Seq=0 Ack=1 Win=0
3	0.967138	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=1 Ack=1 Win=6528
4	0.967862	10.117.220.78	188.184.67.127	HTTP	80	GET / HTTP/1.1
5	1.229785	188.184.67.127	10.117.220.78	TCP	60	80 → 56451 [SYN, ACK] Seq=0 Ack=1 Win=0
6	1.229785	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [FIN, ACK] Seq=197 Ack=18 Win=32
7	1.229785	188.184.67.127	10.117.220.78	HTTP	250	HTTP/1.1 304 Not Modified
8	1.229785	188.184.67.127	10.117.220.78	TCP	54	56451 → 80 [ACK] Seq=1 Ack=1 Win=6528
9	1.230144	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [ACK] Seq=438 Ack=198 Win=0
10	1.230951	10.117.220.78	188.184.67.127	TCP	54	56449 → 80 [FIN, ACK] Seq=438 Ack=198 Win=0
11	1.536721	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [ACK] Seq=198 Ack=439 Win=0
12	1.741871	10.117.220.78	188.184.67.127	TLSv1.2	78	Application Data
13	1.743104	10.117.220.78	188.184.67.127	TCP	54	56449 → 443 [FIN, ACK] Seq=25 Ack=439 Win=0
14	2.070672	188.184.67.127	10.117.220.78	TLSv1.2	78	Application Data
15	2.070672	188.184.67.127	10.117.220.78	TCP	60	443 → 56449 [FIN, ACK] Seq=25 Ack=439 Win=0
16	2.070800	10.117.220.78	188.184.67.127	TCP	54	56447 → 443 [EST, ACK] Seq=26 Ack=25 Win=0
17	2.133370	188.184.67.127	10.117.220.78	TCP	60	443 → 56449 [FIN, ACK] Seq=26 Ack=25 Win=0
18	2.133370	188.184.67.127	10.117.220.78	TCP	54	80 → 56449 [SYN, ACK] Seq=0 Ack=1 Win=0
19	2.565224	10.117.220.78	208.115.231.158	TCP	55	52802 → 443 [ACK] Seq=1 Ack=1 Win=254
20	2.565852	208.115.231.158	10.117.220.78	TCP	54	443 → 52802 [ACK] Seq=0 Ack=2 Win=0
21	2.566088	10.117.220.78	208.115.231.158	TCP	54	[TCP Dup ACK 2594] [TCP ACKed unseen]
22	2.714200	208.115.231.158	10.117.220.78	TCP	60	[TCP Previous segment not captured]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 3046997063

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment Number (raw): 0

1960 ... = Header Length: 32 bytes (8)

Flags: 0x002 (SYN)

...0 ... = Reserved: Not set

...0 ... = Accurate ECN: Not set

...0 ... = Congestion Window Reduced: Not set

...0 ... = ECN-Schoor: Not set

...0 ... = Urgent: Not set

...0 ... = Acknowledgment: Not set

...0 ... = Push: Not set

...0 ... = Reset: Not set

...1 ... = SYN: Set

...0 ... = FIN: Not set

TCP Flags:S.

Window: 65535

[Calculated Window size: 65535]

Checksum: 0x721 [unverified]

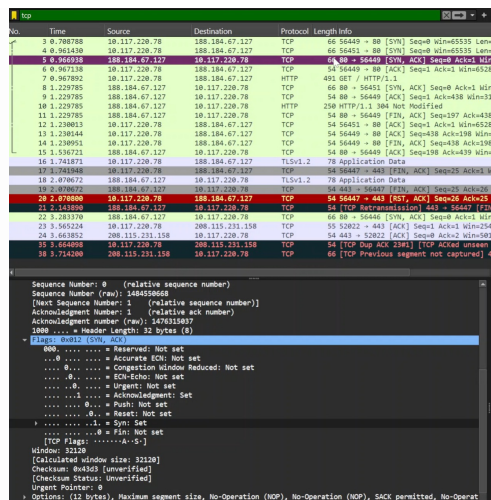
[Checksum Status: Unverified]

Urgent Pointer: 0

Options (12 bytes): Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP)

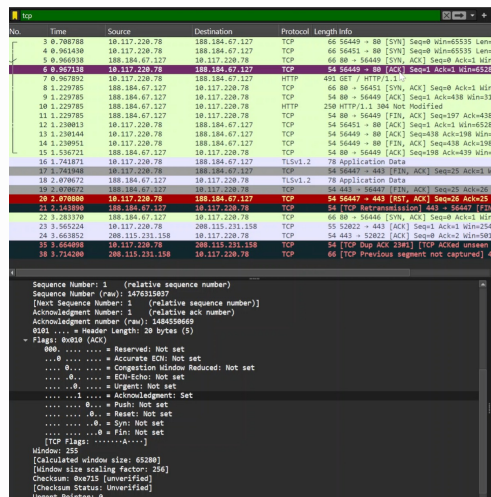
syn

Сервер, как бы, протягивает руку, давая ответ в виде syn и ack, в котором как бы говорит, что теперь знает о клиенте (рис. [-@fig:024]).



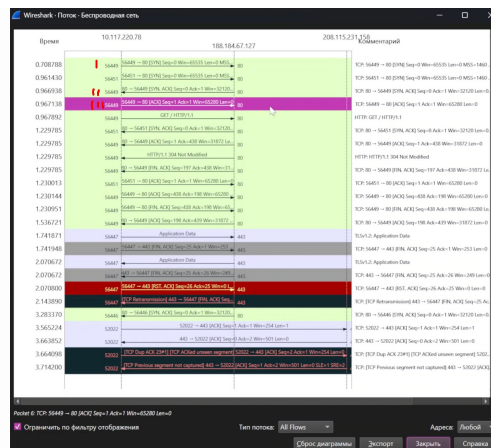
syn + ack

Клиенту остаётся пожать руку, сказав серверу, что тоже теперь о нём знает, при этом флаг syn больше не стоит, поскольку оба они теперь знают о друг друге, и запрос на синхронизацию больше не нужен (рис. [-@fig:025]).



ack

Теперь посмотрим на график потока. Здесь можно явно увидеть процесс хендшейка прямо перед get запросом (рис. [-@fig:026]).



Хендшейк на графике потока

Выводы

В результате выполнения работы были получены навыки анализа пакетов в wireshark